Giorgio Micaletto

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EDUCATION

University of Chicago — **Chicago, IL**
PhD in Statistics — Sep 2025 – Jun 2030 (expected)

- **Research Interests:** Bayesian inference and computation, Monte Carlo methods, geometric methods in statistics
- **Relevant Coursework:** Brownian Motion and Stochastic Calculus, Measure-Theoretic Probability, Asymptotics in Time Series and Stochastic Approximation, Numerical Linear Algebra

Bocconi University — **Milan, IT**
BSc in Economics, Management and Computer Science — Sep 2022 – Jun 2025
Exchange Student, Department of Mathematics, University of Washington — Jan 2025 – Mar 2025

- **Thesis:** *Ricci Curvature, Flow and Graph Convolutional Networks: A Differential-Geometric Framework* (advisor: Prof. Elia Bruè)
- **Relevant Coursework:** Information Geometry, Stochastic Processes (Bocconi); Advanced Probability, PDEs (UW)

PUBLICATIONS & PREPRINTS

- Micaletto, G. (2026). *Overlap, Not Moments: Finite-Sample Theory for Bridge Sampling.* [arXiv:2610.17403]
- Micaletto, G. & Nigrelli, T. (2026). *Scalable Graph Curvature: Bounding Ollivier-Ricci via Local Combinatorics for Geometric Deep Learning.* NeurIPS 2026 Workshop on Graph Distributional Deep Learning. [arXiv:2603.13535]
- Micaletto, G. & Vehtari, A. (2025). *Bridge Sampling Diagnostics.* Revised and resubmitted, *The American Statistician*. [arXiv:2508.14487]

RESEARCH & APPLIED EXPERIENCE

Aalto University — **Helsinki, FI**
Visiting Research Fellow – Department of Computer Science (remote; PI: Prof. Aki Vehtari) — Sep 2024 – Present

- Led the revise-and-resubmit of *Bridge Sampling Diagnostics*: showed the Monte Carlo standard error (MCSE) saturates near 1.05, and validated a 0.3 trust threshold on 41 real posteriors and simulations (95% intervals cover 93–98% below it)
- Designed a hybrid score-matching proposal (affine-invariant midpoint of the sample and score-based covariances) that cut RMSE by 23× to over 1000× on the posteriors where the standard proposal was miscalibrated
- Implemented the MCSE in the bridgesampling R package (merged, CRAN release pending); hybrid proposal in open PRs

Research Intern – Department of Computer Science — Jun 2024 – Aug 2024

- Developed a closed-form MCSE for bridge-sampling evidence estimates (multi-chain ESS correction, exact log-scale variance)
- Contributed to priorsense by automating log-prior evaluation for rstanarm models

Accenture — **Milan, IT**
Research Intern – Data Science and Artificial Intelligence — Jun 2025 – Sep 2025

- Built a vision-LLM pipeline that reconstructs section hierarchies from PDFs without a text layer (13k pages processed) and loads them into a Neo4j knowledge graph behind a personalized RAG assistant

Bocconi University — **Milan, IT**
Research Intern – Department of Decision Sciences (PI: Prof. Omiros Papaspiliopoulos) — Mar 2025 – Jun 2025

- Built the data pipeline for an ML model of theatre ticket sales: scrapers tracking seat occupancy for Athens productions on three ticketing platforms (BeautifulSoup with retries; Selenium for JS seat maps), merged into one schema

Research Intern – Department of Finance (PI: Prof. Marta Zava) — Oct 2023 – Jun 2024

- Rewrote the team's motif-counting code for a project predicting VC funding from founder–investor co-funding networks (2M entries; both one-mode projections, with multi-edges), cutting runtime by 82% versus the original code

SELECTED PROJECTS

- EY AI & Data Challenge (Mar 2026): Semifinalist, top 10 of 953 teams, competing solo. Predicted water quality at South African monitoring stations from coordinates and dates alone, engineering 50+ features from satellite imagery (STAC APIs), terrain and land use; spatial CV with in-fold encoder refits to block leakage, and adversarial validation for train–test shift
- Softmax-Biased Simulated Annealing (Dec 2023): Proved convergence to the global minima under logarithmic cooling via Hajek's theorem, though the proposal breaks Boltzmann detailed balance; mapped the basins with zeroth Betti numbers of sublevel sets

AWARDS

- Best Poster Award (of 120), Aalto Science Institute summer research programme, for *Bridge Sampling Diagnostics* (2024)
- 1st of 12 finalists, Bocconi internal programming contest; represented Bocconi at ITACPC and ICPC SWERC (2024–25 season)
- Full-ride merit scholarship, Bocconi University (2022–2025), incl. exchange funding for UW (awarded 2024)
- Thesis Fair speaker (2025): shortlisted by the program director (top 7%) and voted one of six presenters by students
- Best Student Lecturer of the cohort, Applied Stochastic Processes (2025)

ACADEMIC SERVICE & LEADERSHIP

- Incoming Editorial Assistant, *Biometrika*
- PhD Student Representative, University of Chicago (2026–present)
- Student Council member; class and course representative, Bocconi University (2022–2025)

SKILLS

- **Programming:** Python (PyTorch, scikit-learn, XGBoost, pandas, Selenium), R (Stan, rstanarm, package development), SQL
- **Tools:** Git, Docker, Linux, Slurm, Neo4j, LaTeX; **Languages:** Italian (native), English (fluent)